

BRIGHT LEARN EXERCISE 2 - SQL AGGREGATE FUNCTIONS & OPERATIONS

01 The Students Table
students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT Finance
4	Diana	23	Finance
5	Eve	22	HR

01) SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

02) SELECT department, AVG (age) AS avg-age
FROM students
GROUP BY department;

department	avg-age
IT	20.5
HR	22
Finance	23

03) SELECT department, COUNT(student_id) AS student_count
 FROM students
 GROUP BY department
 HAVING COUNT(student_id) > 1;

department	student_count
IT	2
HR	2

04) SELECT student_id,
 name,
 department
 FROM students
 WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05) SELECT student_id,
 name,
 age,
 department
 FROM students
 WHERE department = 'IT' OR 'HR' AND age > 21;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
5	Eve	22	HR

02. The courses Table

Courses

course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

06) SELECT department,
 sum(credits) AS total_credits
FROM courses
GROUP BY department;

department	total_credits
IT	11
HR	2
Finance	3

07) SELECT course_id,
 course_name,
 department
FROM courses
WHERE ~~NOT credits~~ credits NOT BE NOT credits = 4

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

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08) SELECT course_id,
        course_name,
        credits
FROM     courses
ORDER BY credits DESC
LIMIT   3;

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course-id	course-name	credits
102	Python	4
103	Data science	4
101	SQL Basics	3

03 The enrollments Table

enrollments

enrollment-id	student-id	course-id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

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09) SELECT MAX(grade) AS max-grade,
        MIN (grade) AS min-grade,
        AVG (grade) AS avg-grade
FROM     enrollments;

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max-grade	min-grade	avg-grade
90	78	84.6

i6) SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 The salaries Table

salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

11) SELECT department,
 SUM(salary) AS total_salary,
 SUM(bonus) AS total_bonus
 FROM salaries
 GROUP BY department;

department	total_salary	total_bonus
IT	122000	11500
HR	109000	7500
Finance	70000	6000

12) SELECT department, AVG(salary) AS avg_salary
 FROM salaries
 GROUP BY department
 HAVING AVG(salary) > 55000;

department	avg_salary
IT	61000
HR	54500
Finance	70000

13) SELECT employee-id, name,
 (salary+bonus) AS total-compensation
 FROM salaries
 WHERE (salary+bonus) > 60000

employee-id	name	total-compensation
1	Tom	65000
3	Spike	76000
4	Tyke	67500

15 The projects Table

Projects

project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

14) SELECT department,
 sum(budget) AS total-budget^d
 sum(budget) AS total-avg-budget
 FROM projects
 GROUP BY department
 HAVING AVG(budget) > 70000

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

15) SELECT project-id,
project-name,
department,
budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND
department NOT IN ('Marketing')

project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000